

Deep Learning-Based 3D Reconstruction for Personalized Orthotic Models

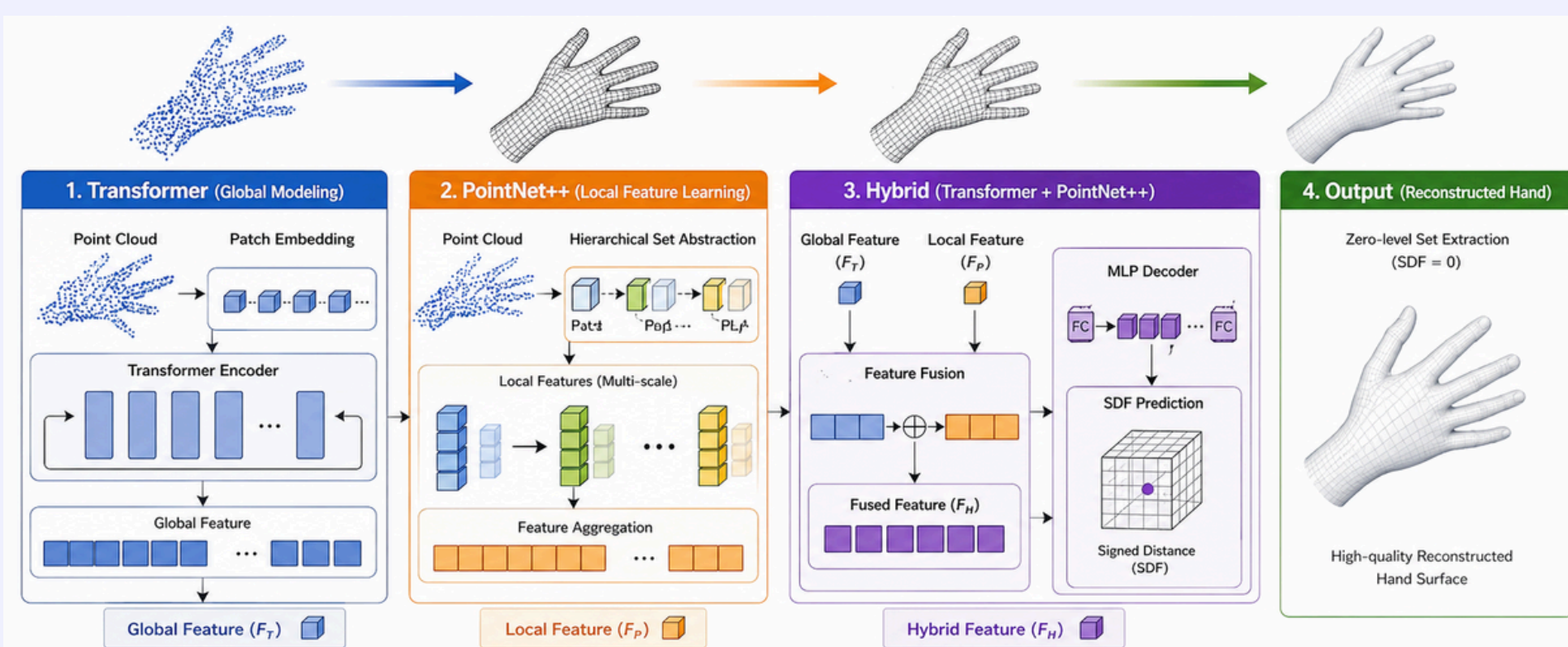
Sumaia Abdulwahab Alwahashi ID: 461217096 Supervisor: Dr,Aicha Ben Makhoulouf
Department AI of Computer Science, College of Computer

Introduction

Deep learning techniques have been widely developed to process 3D data directly from point clouds. It presents a comparative analysis of PointNet++, a Transformer-based reconstruction model, and a proposed hybrid framework that integrates both approaches for 3D reconstruction from imperfect point cloud data. The evaluation is conducted based on geometric accuracy, structural detail preservation, and surface quality. These findings highlight the complementary nature of local and global representations and support the selection of hybrid strategies for accurate and scalable medical 3D reconstruction.

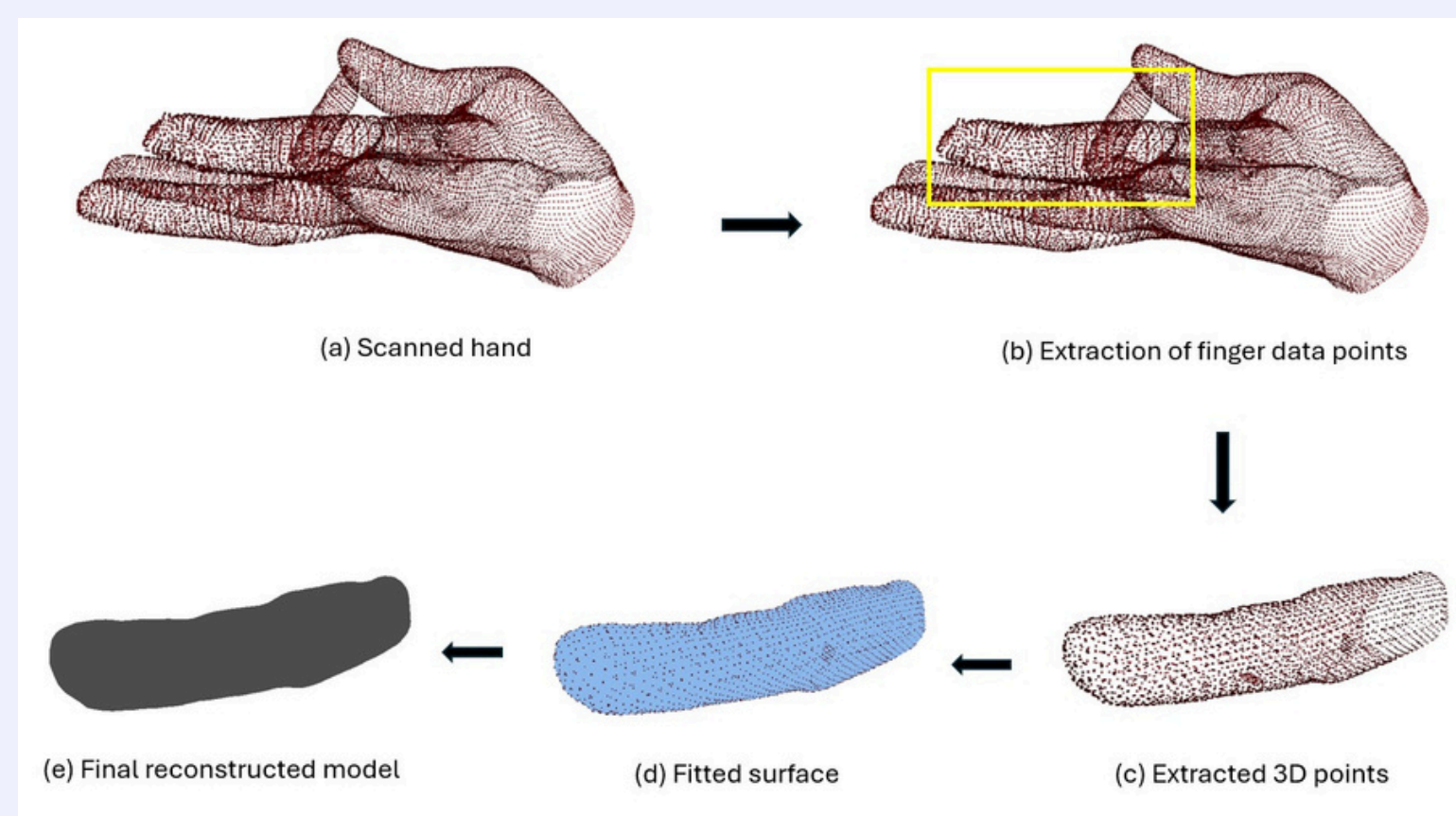
Methodology

This work follows a step-by-step flow: data collection, data pre-processing, training three deep learning models, and final 3D reconstruction. Each of these models receives a sparse input point cloud to reconstruct a denser surface representation that accurately captures the geometry of the original surface.



Results

These results validate the architectural hypothesis motivating the Hybrid design combining local geometric feature learning PointNet++ branch with global shape context modelling Transformer branch through bidirectional crossattention, feature-weighted seed selection.



Comparison Table

Metric	Transformer	PointNet++	Hybrid
Chamfer Dist.	0.0006836	0.0006995	0.0006488
Precision	0.9936	0.9954	0.9958
Recall	0.9986	0.9983	0.9993
F-Score	0.9960	0.9969	0.9975
Normal Consist.	0.9893	0.9802	0.9922
Hausdorff Dist.	0.0681	0.0687	0.0638

Models

- Transformer**
captures global relationships and context
- PointNet++**
strong local feature extraction
- Hybrid model**
combines local and global feature learning

Conclusion

- ✓ Experimental results show that the hybrid approach achieves the best performance across all evaluation metrics, including Chamfer Distance, Hausdorff Distance, Precision, Recall, and F-score.
- ✓ The model produces more accurate, complete, and geometrically consistent 3D hand reconstructions compared to individual models.
- ✓ These findings demonstrate that combining local and global representations is an effective strategy for robust and high-quality 3D reconstruction from point cloud data.